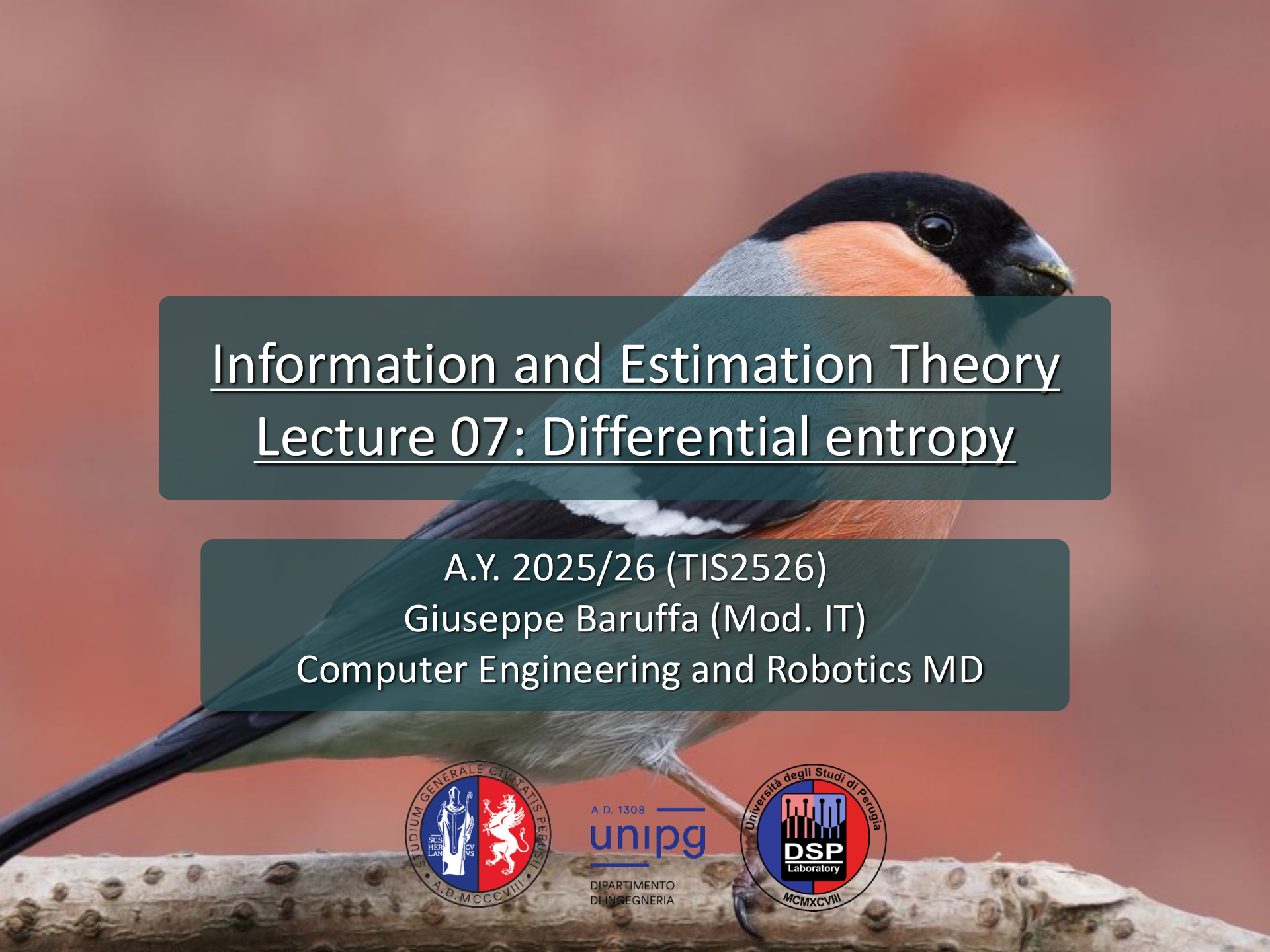


Information and Estimation Theory
Lecture 07: Differential entropy

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Continuous RVs and Differential Entropy

- Differential entropy is defined as the entropy of a continuous RV
- It is also related to the shortest description length and is similar to the entropy of a discrete RV
  - There are some important differences, and some care is needed

- Let X be a RV with **cumulative distribution function (CDF)** $F(x) = \Pr\{X \leq x\}$
  - If $F(x)$ is continuous, the RV is **continuous**
- Let $f(x) = F'(x)$, when the derivative is defined
- If $\int_{-\infty}^{\infty} f(x)\mathrm{d}x = 1$, $f(x)$ is called the **probability density function (pdf)** for X
- The set $\mathcal{S} = \{x: f(x) > 0\}$ is called the **support set** of X

Differential Entropy

- The **differential entropy (DE)** $h(X)$ of a continuous RV X with pdf $f(x)$ is defined as

$$h(X) = -\int_S f(x) \log f(x)\, dx = \mathrm{E}\{-\log f(x)\}$$

  where $\mathcal{X}$ is the support of X

- As in the discrete case, the DE depends only on the pdf of the RV, and sometimes the DE is written as $h(f)$ rather than $h(X)$

- As in every example involving an integral, or even a pdf, we should include the statement "if it exists"
  - There are examples of RVs for which a pdf function does not exist or for which the above integral does not exist
    - E.g., continuous-discrete mixed distributions

Example: Uniform PDF

- Consider a RV distributed uniformly from 0 to $a > 0$ so that its pdf is $1/a$ from 0 to a and 0 elsewhere
- Then its DE is

$$h(X) = -\int_0^a \frac{1}{a} \log \frac{1}{a} dx = \frac{1}{a} \log a \int_0^a dx = \log a$$

  - E.g., a uniform variate in [0,1) "informs" with 0 bits/symbol
- Note: For $a < 1$, $\log(a) < 0$, and the DE is negative
  - Hence, unlike discrete entropy, DE can be negative
- However, $2^{h(X)} = 2^{\log a} = a$ is the **volume** of the support set, which is always nonnegative, as we expect

Example: Normal PDF

- Let $X \sim \varphi(x) = \mathcal{N}(0,\sigma^2) = 1/\sqrt{(2\pi\sigma^2)}e^{-x^2/(2\sigma^2)}$. Then the DE in *nats* is

$$h_e(\phi) = -\int_{-\infty}^{+\infty} \phi \ln \phi \, dx = -\int_{-\infty}^{+\infty} \phi(x) \left[-\frac{x^2}{2\sigma^2} - \ln\sqrt{2\pi\sigma^2} \right] dx$$

$$= \frac{\mathrm{E}\{X^2\}}{2\sigma^2} + \frac{1}{2}\ln 2\pi\sigma^2 = \frac{1}{2} + \frac{1}{2}\ln 2\pi\sigma^2 = \frac{1}{2}\ln e + \frac{1}{2}\ln 2\pi\sigma^2$$

$$= \frac{1}{2}\ln 2\pi e\sigma^2 \qquad \text{[nats]}$$

- Or in bits

$$h(\phi) = h_2(\phi) = \log_2 e \, h_e(\phi) = \frac{1}{2}\log 2\pi e\sigma^2 \qquad \text{[bits]}$$

  - E.g., a Gaussian with unit variance "informs" with 2.0471 bits/symbol

AEP of Continuous RV

- An important role of discrete RVs entropy was in the AEP
  - For a sequence of i.i.d. RVs, $p(X_1,X_2,\ldots,X_n)$ is close to $2^{-nH(X)}$ with high prob.
- This enabled us to define the typical set and characterize the behavior of typical sequences → do the same for continuous RVs

- **Theorem:** Let $X^n=X_1,X_2,\ldots,X_n$ be a sequence of RVs i.i.d. $\sim f(x)$. Then

$$-\frac{1}{n}\log f(X^n) \xrightarrow{\mathrm{P}} \mathrm{E}\{-\log f(X)\} = h(X) \qquad \text{in probability}$$

  - *Proof:* The proof follows directly from the weak law of large numbers

Typical Set

- For $\varepsilon > 0$ and any n, we define the typical set $A_\varepsilon^{(n)}$ w.r.t. $f(x)$ as

$$A_\epsilon^{(n)} = \left\{ x^n \in \mathcal{S}^n : \left| -\frac{1}{n} \log f(x^n) - h(X) \right| \leq \epsilon \right\}$$

  where $f(x^n) = \prod_{i=1}^{n} f(x_i)$

- The properties of the typical set for continuous RVs match those for discrete RVs
- The analog of the cardinality of the typical set for the discrete case is the volume of the typical set for continuous RVs

- The volume **Vol($\mathcal{A}$)** of a set $\mathcal{A} \subset \mathcal{X}^n$ is defined as

$$\mathrm{Vol}(\mathcal{A}) = \int_{\mathcal{A}} dx_1 dx_2 \ldots dx_n$$

Theorem of Typical Set

- **Theorem:** The typical set $A_\varepsilon^{(n)}$ has the following properties:
  1. $\Pr\{A_\varepsilon^{(n)}\} > 1 - \varepsilon$ for n sufficiently large
  2. $\text{Vol}(A_\varepsilon^{(n)}) \leq 2^{n(h(X)+\varepsilon)}$ for all n
  3. $\text{Vol}(A_\varepsilon^{(n)}) \geq (1-\varepsilon)2^{n(h(X)-\varepsilon)}$ for n sufficiently large

  - *Proof:* for property 1, use the previous theorem

$$-\frac{1}{n}\log f(X^n) = -\frac{1}{n}\log\prod_{i=1}^n f(X_i) = -\frac{1}{n}\sum_{i=1}^n \log f(X_i) \to h(X) \quad \text{and for } n > n_0$$

$$\Pr\left\{\left|-\frac{1}{n}\sum_{i=1}^n \log f(X_i) - h(X)\right| < \epsilon\right\} > 1-\delta$$

  - For property 2

$$1 = \int_{S^n} f(x_1, x_2, \ldots, x_n)dx_1dx_2\ldots dx_n \geq \int_{A_\epsilon^{(n)}} f(x_1, x_2, \ldots, x_n)dx_1dx_2\ldots dx_n$$

From definition of typical set

$$\geq \int_{A_\epsilon^{(n)}} 2^{-n(h(X)+\epsilon)}dx_1dx_2\ldots dx_n = 2^{-n(h(X)+\epsilon)}\int_{A_\epsilon^{(n)}} dx_1dx_2\ldots dx_n$$

$$= 2^{-n(h(X)+\epsilon)}\text{Vol}\left(A_\epsilon^{(n)}\right)$$

    - ❖ We argue that the volume of the typical set is at least this large

Theorem of Typical Set (proof cont'd)

- If *n* is sufficiently large so that property 1 is satisfied, then comes property 3

$$1-\epsilon=\int_{A_\epsilon^{(n)}} f(x_1,x_2,\ldots,x_n)dx_1dx_2\ldots dx_n \le \int_{A_\epsilon^{(n)}} 2^{-n(h(X)-\epsilon)}dx_1dx_2\ldots dx_n$$
$$=2^{-n(h(X)-\epsilon)}\int_{A_\epsilon^{(n)}} dx_1dx_2\ldots dx_n = 2^{-n(h(X)-\epsilon)}\mathrm{Vol}\left(A_\epsilon^{(n)}\right)$$

- Thus, for *n* sufficiently large, we have

$$(1-\epsilon)2^{n(h(X)-\epsilon)} \le \mathrm{Vol}\left(A_\epsilon^{(n)}\right) \le 2^{n(h(X)+\epsilon)}$$

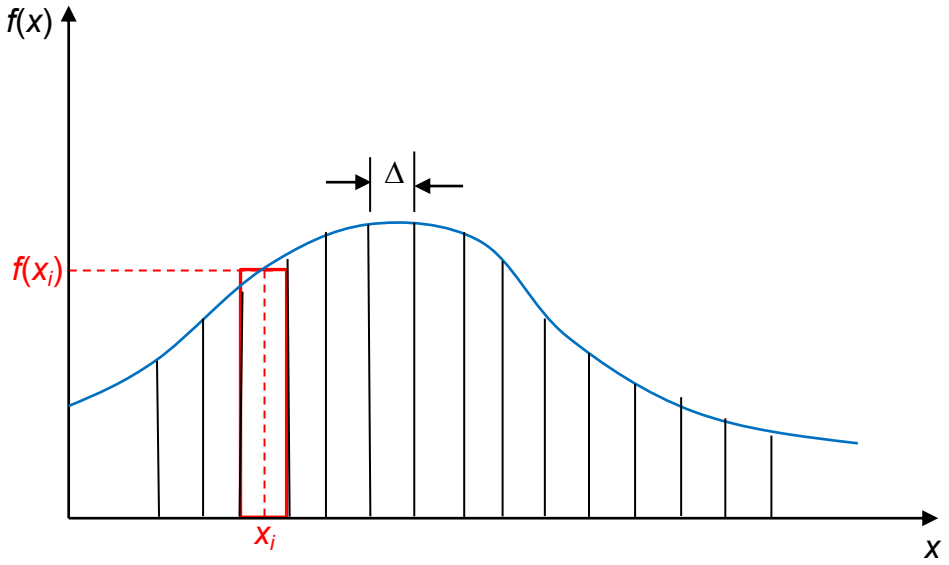
Smallest Volume Set Theorem

- **Theorem:** The set $A_\varepsilon^{(n)}$ is the smallest volume set with probability $\geq 1-\varepsilon$, to first order in the exponent
  - *Proof:* Same as in the discrete case

- So, the **volume** of the smallest set that contains most of the probability **is approximately 2^{nh}**
- This is an n-dim. volume, so the side length is $(2^{nh})^{1/n} = 2^h$
- An interpretation of DE: <u>the logarithm of the equivalent side length of the smallest set that contains most of the probability</u>
  - Low entropy → the RV is confined to a small effective volume
  - High entropy → the RV is widely dispersed

Relation between DE and Entropy

- Consider a RV $X \sim f(x)$
- Divide the range of X in Δ-wide bins
  - Assume continuous pdf within the bins
- Then, by the mean value theorem, there exists a value x_i within each bin such that



$$f(x_i)\Delta = \int_{i\Delta}^{(i+1)\Delta} f(x)dx$$

- Consider the quantized RV $[X]_\Delta = x_i$ if $i\Delta \leq X < (i+1)\Delta$
- Then the probability that $[X]_\Delta = x_i$ is

$$p_i = \int_{i\Delta}^{(i+1)\Delta} f(x)dx = f(x_i)\Delta$$

Relation between DE and Entropy

- The (discrete) entropy of the quantized version is

$$H([X]_\Delta) = -\sum_{i=-\infty}^{+\infty} p_i \log p_i = -\sum_{i=-\infty}^{+\infty} f(x_i)\Delta \log(f(x_i)\Delta)$$

$$= -\sum_{i=-\infty}^{+\infty} \Delta f(x_i) \log f(x_i) - \sum_{i=-\infty}^{+\infty} f(x_i)\Delta \log \Delta$$

$$= \boxed{-\sum_{i=-\infty}^{+\infty} \Delta f(x_i) \log f(x_i)} - \log \Delta$$

since $\Sigma_i f(x_i)\Delta = \int f(x)dx = 1$

- If $f(x) \log f(x)$ is Riemann-integrable (so that the limit is well defined), the **1st term** approaches the integral of $-f(x) \log f(x)$ as $\Delta \to 0$ (by definition of Riemann integrability)

$$\int_a^b g dx = \int_a^b \lim_{n\to\infty} g_n \, dx = \lim_{n\to\infty} \int_a^b g dx$$

Entropy of Quantized Continuous RV

- **Theorem:** If the pdf $f(x)$ of the RV X is Riemann-integrable, then

$$\lim_{\Delta \to 0} (H([X]_\Delta) + \log \Delta) = h(f) = h(X)$$

Thus, the discrete entropy of an n-bit quantization of a continuous RV X is <u>approximately</u>

$$H([X]_\Delta) \approx h(X) + n$$

  - *Proof:* see before. Also, $\Delta = 2^{-n}$ is the resolution or number of fractional bits required to represent the number x

- $X \sim \mathcal{U}[0,1]$ and $\Delta = 2^{-n}$
  - $H([X]_\Delta) = h(X) + n = 0 + n$
- $X \sim \mathcal{U}[0,⅛]$, the first 3 fractional bits are 0
  - Agrees with $H([X]_\Delta) = n - 3$

Entropy of Quantized Continuous RV

- $h(X) + n$ is the <u>average</u> number of information bits required to describe the RV X to n bit accuracy
  - A Gaussian RV requires 1-bit accuracy to achieve a discrete entropy of 1 bit if $\sigma = 0.2420$

$$1 = \frac{1}{2}\log 2\pi e\sigma^2 + 1 \qquad \frac{1}{2}\log 2\pi e\sigma^2 = 0 \qquad \log 2\pi e\sigma^2 = \log 1 \qquad \sigma = \frac{1}{\sqrt{2\pi e}}$$

  - $X \sim \mathcal{N}(0,100)$ truncated at n-bit precision achieves $n + \frac{1}{2}\log(2\pi e 100) = n + 5.37$ bits of discrete entropy

- For a discrete RV, $H([X]_\Delta) = H(X)$ under fine quantization, i.e., when $\Delta \to 0$
- Then $h(X) = H(X) + \log \Delta = -\infty$ and Vol(S)=$2^{-\infty}$=0, the volume of the support set is null

Joint and Conditional DE

- The **joint DE** of a set $X_1,X_2,\ldots,X_n$ of RVs with pdf $f(x_1,x_2,\ldots,x_n)$ is

$$h(X^n) = -\int f(x^n)\log f(x^n)\,dx^n$$

- If X,Y have a joint pdf $f(x,y)$, we define the conditional DE

$$h(X|Y) = -\int f(x,y)\log f(x|y)\,dxdy$$

- Since in general $f(x|y) = f(x,y)/f(y)$, we can also write

$$h(X|Y) = h(X,Y) - h(Y)$$

  - Be careful if any of the DEs are infinite

Entropy of a Multivariate Normal Distribution

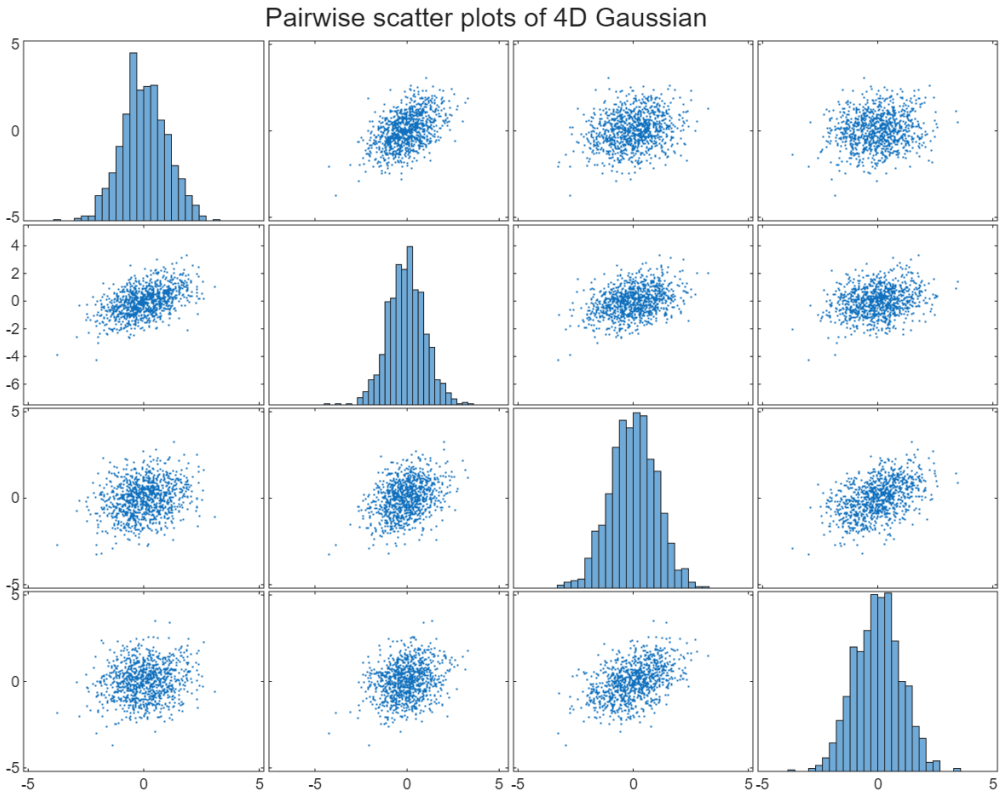
- **Theorem:** Let $\mathbf{X} = (X_1, X_2, \ldots, X_n)^T$ have a multivariate normal pdf with mean $\boldsymbol{\mu} = E\{\mathbf{X}\}$ and covariance matrix $\mathbf{K} = E\{(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^T\}$. Then

$$h(\mathbf{X}) = h\left(\mathcal{N}_n(\boldsymbol{\mu}, \mathbf{K})\right) = \frac{1}{2}\log\left((2\pi e)^n |\mathbf{K}|\right) \quad \text{bits}$$

where $|\mathbf{K}|$ denotes the determinant of $\mathbf{K}$

  - *Proof:* The pdf of $X_1, X_2, \ldots, X_n$ is

$$f(\mathbf{x}) = \frac{1}{\left(\sqrt{2\pi}\right)^n |\mathbf{K}|^{\frac{1}{2}}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \mathbf{K}^{-1}(\mathbf{x}-\boldsymbol{\mu})}$$



Proof cont'd

$$h_e(f) = -\int f(\mathbf{x})\left[-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\mathbf{K}^{-1}(\mathbf{x}-\boldsymbol{\mu}) - \ln\left(\left(\sqrt{2\pi}\right)^n|\mathbf{K}|^{\frac{1}{2}}\right)\right]d\mathbf{x}$$

$$= \frac{1}{2}\mathrm{E}\left\{\sum_i\sum_j (X_i-\mu_i)(\mathbf{K}^{-1})_{ij}(X_j-\mu_j)\right\} + \frac{1}{2}\ln\left((2\pi)^n|\mathbf{K}|\right)$$

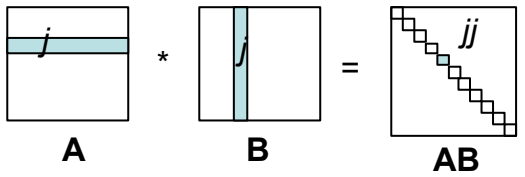
$$= \frac{1}{2}\mathrm{E}\left\{\sum_i\sum_j (X_i-\mu_i)(X_j-\mu_j)(\mathbf{K}^{-1})_{ij}\right\} + \frac{1}{2}\ln\left((2\pi)^n|\mathbf{K}|\right)$$

$$= \frac{1}{2}\sum_i\sum_j \mathrm{E}\{(X_j-\mu_j)(X_i-\mu_i)\}(\mathbf{K}^{-1})_{ij} + \frac{1}{2}\ln\left((2\pi)^n|\mathbf{K}|\right)$$

row1xcol1 + row2xcol2 + ...

$$= \frac{1}{2}\sum_j\sum_i \mathbf{K}_{ji}(\mathbf{K}^{-1})_{ij} + \frac{1}{2}\ln\left((2\pi)^n|\mathbf{K}|\right)$$

...= sum along diagonal of product



$$= \frac{1}{2}\sum_j (\mathbf{K}\mathbf{K}^{-1})_{jj} + \frac{1}{2}\ln\left((2\pi)^n|\mathbf{K}|\right)$$

$$= \frac{1}{2}\sum_j \mathbf{I}_{jj} + \frac{1}{2}\ln\left((2\pi)^n|\mathbf{K}|\right) = \frac{n}{2} + \frac{1}{2}\ln\left((2\pi)^n|\mathbf{K}|\right) = \frac{1}{2}\ln\left((2\pi e)^n|\mathbf{K}|\right) \text{ nats}$$

$$h(f) = \frac{1}{2}\log\left((2\pi e)^n|\mathbf{K}|\right) \quad \text{bits}$$

Relative Entropy and Mutual Information

- The **relative entropy** (or **Kullback–Leibler distance**) $D(f \| g)$ between two pdfs f and g is

$$D(f \parallel g) = \int_{x \in S_f} f(x) \log \frac{f(x)}{g(x)} dx$$

g must be >0 where f is >0

- Note that $D(f \| g)$ is finite only if the support set of f is contained in the support set of g
  - Motivated by continuity, we set 0 log(0/0) = 0

- The **mutual information** $I(X;Y)$ between two RVs with joint pdf $f(x,y)$ is

$$I(X;Y) = \int_{x,y \in S_f} f(x,y) \log \frac{f(x,y)}{f(x)f(y)} dxdy$$

$$I(X;Y) = h(X) - h(X|Y) = h(Y) - h(Y|X) = h(X) + h(Y) - h(X,Y)$$

$$I(X;Y) = D\big(f(x,y) \parallel f(x)f(y)\big)$$

Relative Entropy and Mutual Information

- The properties of $D(f\|g)$ and $I(X;Y)$ are like in the discrete case
- In particular, the mutual information between two RVs is the limit of the mutual information between their quantized versions

$$I([X]_\Delta;[Y]_\Delta) = H([X]_\Delta) - H([X]_\Delta|[Y]_\Delta)$$
$$\approx h(X) - \log\Delta - (h(X|Y) - \log\Delta) = I(X;Y)$$

- More generally, we can define mutual information in terms of finite partitions of the range of the RV
- Let $\mathcal{S}_X$ be the range of a RV X. A **partition** $\mathcal{P}$ of $\mathcal{S}_X$ is a finite collection of disjoint sets P_i such that $\bigcup_i P_i = \mathcal{S}_X$
- The quantization of $\mathcal{X}$ by $\mathcal{P}$ (denoted $[X]_\mathcal{P}$) is the discrete RV

$$\Pr\{[X]_\mathcal{P} = i\} = \Pr\{X \in P_i\} = \int_{P_i} dF(x)$$

Relative Entropy and Mutual Information

- For two RVs *X* and *Y* with partitions $\mathcal{P}$ and $\mathcal{Q}$, we can calculate the mutual information between the quantized versions of *X* and *Y*
- The **mutual information** between two RVs *X* and *Y* is given by

$$I(X;Y) = \sup_{\mathcal{P},\mathcal{Q}} I\left([X]_{\mathcal{P}};[Y]_{\mathcal{Q}}\right)$$

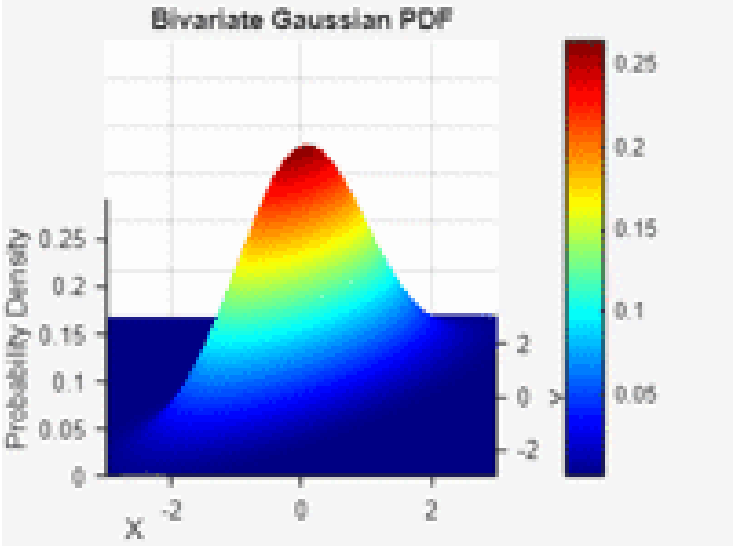
By definition, a max must be in the set, while a sup can be just outside

where the supremum is over all finite partitions $\mathcal{P}$ and $\mathcal{Q}$

- This is the "master" definition of mutual information that always applies to joint pdfs with point masses (atoms), densities, and singular parts
- Moreover, by continuing to refine the partitions $\mathcal{P}$ and $\mathcal{Q}$, one finds a monotonically increasing sequence $I([X]_{\mathcal{P}};[Y]_{\mathcal{Q}}) \nearrow I(X;Y)$

Limit approaching from the left

Example



- Mutual information between correlated Gaussian RVs with correlation ρ
- Let $(X,Y) \sim \mathcal{N}_2(0,\mathbf{K})$, where

$$\mathbf{K} = \begin{bmatrix} E\{X^2\} & E\{XY\} \\ E\{YX\} & E\{Y^2\} \end{bmatrix} = \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix}$$

- Then $h(X) = h(Y) = \frac{1}{2}\log(2\pi e\sigma^2)$ and $h(X,Y) = \frac{1}{2}\log\left((2\pi e)^2|\mathbf{K}|\right) = \frac{1}{2}\log\left((2\pi e)^2\sigma^4(1-\rho^2)\right)$, thus

$$I(X;Y) = h(X) + h(Y) - h(X,Y) = \frac{1}{2}\log\left((2\pi e)^2\sigma^4\right)$$
$$-\frac{1}{2}\log\left((2\pi e)^2\sigma^4(1-\rho^2)\right) = -\frac{1}{2}\log(1-\rho^2)$$

  - If $\rho = 0$, X and Y are independent, and the mutual information is 0
  - If $\rho \to \pm 1$, X and Y are perfectly correlated, and $I(X;Y) \to \infty$
  - Mutual information does not depend on marginal variance σ^2

Properties

- **Theorem:** $D(f\|g) \geq 0$ with equality iff $f = g$ almost everywhere
  - *Proof:* Let $\mathcal{S}$ be the support set of f. Then

$$-D(f \parallel g) = \int_{\mathcal{S}} f(x) \log \frac{g(x)}{f(x)} dx \leq \log \int_{\mathcal{S}} f(x) \frac{g(x)}{f(x)} dx = \log \int_{\mathcal{S}} g(x) dx \leq \log 1 = 0$$

E{log(g/f)} ≤ log(E{g/f}) since log is concave (-convex)

We have equality iff we have equality in Jensen's inequality, which occurs iff $f = g$ a.e. a.e. = everywhere except in a subset of volume 0

- **Corollary:** $I(X;Y) \geq 0$ with equality iff X and Y are independent

- **Corollary:** $h(X|Y) \leq h(X)$ with equality iff X and Y are independent

Properties

- **Theorem:** Chain rule for DE

$$h(X_1, X_2, \ldots, X_n) = \sum_{i=1}^{n} h(X_i | X_1, X_2, \ldots, X_{i-1})$$

  - *Proof:* Follows directly from the definitions of joint and conditional pdfs

- **Corollary:** $h(X_1,X_2,\ldots,X_n) \leq \Sigma_i\, h(X_i)$ with equality iff $X_1,X_2,\ldots,X_n$ are independent
  - *Proof:* Follows directly from $h(X_i|X_j,\ldots) \leq h(X_i)$

Hadamard's Inequality

- If we let $\mathbf{X} \sim \mathcal{N}_n(0,\mathbf{K})$ be a multivariate normal RV, calculating the entropy in the previous inequality gives us the **Hadamard's inequality**

$$|\mathbf{K}| \leq \prod_{i=1}^{n} \mathbf{K}_{ii}$$

Determinant of the covariance matrix is upper bounded by the product of the variances

$$\frac{1}{2}\log\left((2\pi e)^n|\mathbf{K}|\right) \leq \sum_{i=1}^{n}\frac{1}{2}\log\left(2\pi e\sigma_i^2\right) \qquad \log\left((2\pi e)^n|\mathbf{K}|\right) \leq \log\left((2\pi e)^n\prod_{i=1}^{n}\sigma_i^2\right)$$

- Several determinant-based inequalities can be derived in this fashion from information-theoretic inequalities
- Note that $\mathbf{K}$ is Hermitian, or $\mathbf{K} = \mathbf{K}^H = (\mathbf{K}^*)^T$

Properties under Affine Transformations

| | stands for absolute value, here and below

- **Theorem:** if $a \neq 0$, then $h(aX+b) = h(X) + \log|a|$

  - *Proof:* Let $Y=aX+b$, then $x=(y-b)/a$. Use

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

$$f_Y(y) = \begin{cases} \left|\frac{d}{dy} g^{-1}(y)\right| f_X\left(g^{-1}(y)\right) & \text{if } y = g(x) \text{ for some } x \\ 0 & \text{otherwise.} \end{cases}$$

  in the DE, then change variable

$$h(Y) = -\int_{\mathcal{S}_Y} f_Y(y) \log f_Y(y)\, dy = -\int_{\mathcal{S}_Y} \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right) \log\left(\frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)\right) dy$$

x=(y-b)/a x=(y-b)/a dy = |a| dx

$$= -\int_{\mathcal{S}_X} f_X(x) \log f_X(x)\, dx + \int_{\mathcal{S}_X} f_X(x) \log|a|\, dx = h(X) + \log|a|$$

Properties under Affine Transformations

- **Corollary:** Translation does not change DE, $h(X + c) = h(X)$

- **Corollary:** $h(\mathbf{AX}) = h(\mathbf{X}) + \log|\det(\mathbf{A})|$

  Linear transformation of RVs **A** *here is a square full-rank matrix, so that*

$$E\{\mathbf{YY}^T\} = \mathbf{A}E\{\mathbf{XX}^T\}\mathbf{A}^T$$

$$f_{Y_1,Y_2}(y_1, y_2) = \frac{1}{|\det(\mathbf{A})|} f_{X_1,X_2}\left(\mathbf{A}^{-1}[\mathbf{y} - \mathbf{b}]\right)$$

Theorem on Multivariate Normal as DE Bound

- **Theorem:** The multivariate normal pdf maximizes the entropy over all pdfs with the same covariance

  Let RV $\mathbf{X} \in \mathbb{R}^n$ with 0 mean and covariance $\mathbf{K} = E\{\mathbf{X}\mathbf{X}^T\}$, i.e., $\mathbf{K}_{ij} = E\{X_i X_j\}$, $1 \le i,j \le n$. Then

  same covariance of multivariate normal

  $$h(\mathbf{X}) \le (1/2)\log[(2\pi e)^n |\mathbf{K}|], \text{ with eq. iff } \mathbf{X} \sim \mathcal{N}_n(0,\mathbf{K})$$

  - *Proof:* Let $g(\mathbf{x})$ be any pdf satisfying $\int g(\mathbf{x}) x_i x_j \, d\mathbf{x} = \mathbf{K}_{ij}$ $\forall i,j$. Let $\varphi_{\mathbf{K}}$ be the pdf of a $\mathcal{N}(0,\mathbf{K})$ random vector with $\boldsymbol{\mu} = 0$.

    Has the same covariance of the multinormal

$$0 \le D(g \parallel \phi_{\mathbf{K}}) = \int_{\mathcal{S}_{\mathbf{x}}} g(\mathbf{x}) \ln \frac{g(\mathbf{x})}{\phi_{\mathbf{K}}(\mathbf{x})} d\mathbf{x} = -h(g) - \int_{\mathcal{S}_{\mathbf{x}}} g(\mathbf{x}) \ln \phi_{\mathbf{K}}(\mathbf{x}) \, d\mathbf{x}$$

Cross-entropy

See Bates (Ch02) on quadratic forms

$$= -h(g) - \mathrm{E}_g\{\log \phi_{\mathbf{K}}(\mathbf{X})\} = -h(g) - \mathrm{E}_g \left\{ \ln \frac{1}{(\sqrt{2\pi})^n |\mathbf{K}|^{\frac{1}{2}}} e^{-\frac{1}{2}\mathbf{X}^T \mathbf{K}^{-1} \mathbf{X}} \right\}$$

$\mathrm{E}_g\{\mathbf{X}^T\mathbf{K}^{-1}\mathbf{X}\}$
$= \mathrm{tr}(\mathbf{K}^{-1}\mathbf{K}) = n$
$= E_{\phi_{\mathbf{K}}}\{\mathbf{X}^T\mathbf{K}^{-1}\mathbf{X}\}$

$$= -h(g) + \ln(\sqrt{2\pi})^n |\mathbf{K}|^{\frac{1}{2}} + \frac{1}{2}\mathrm{E}_g\{\mathbf{X}^T\mathbf{K}^{-1}\mathbf{X}\}$$

$$= -h(g) + \ln(\sqrt{2\pi})^n |\mathbf{K}|^{\frac{1}{2}} + \frac{1}{2}\mathrm{E}_{\phi_{\mathbf{K}}}\{\mathbf{X}^T\mathbf{K}^{-1}\mathbf{X}\}$$

$$= -h(g) - \mathrm{E}_{\phi_{\mathbf{K}}}\{\log \phi_{\mathbf{K}}(\mathbf{X})\} = -h(g) + h(\phi_{\mathbf{K}})$$

Theorem on Multivariate Normal as DE Bound

- Then $h(g) \leq h(\phi_{\mathbf{K}})$

- For the folks: the Gaussian distribution maximizes the entropy over all distributions with the same covariance matrix

- This means that the maximum uncertainty in data is achieved when they tend to have (naturally or artificially induced) a normal distribution, keeping their (cross)variance unchanged

- Any non-Gaussian structure can (potentially) be exploited for better modeling or to achieve compression

Estimation Error and Differential Entropy

- **Theorem (Fano for cont. RV):** For any RV X and constant estimator $\hat{X}$,

$$\mathrm{E}\left\{\left(X-\hat{X}\right)^2\right\} \geq \frac{1}{2\pi e} e^{2h(X)}$$

with equality iff X is Gaussian and $\hat{X}$ is the mean of X

  - *Proof:*

Use nats entropy as upper bound $h(X) \leq (1/2)\ln(2\pi e\sigma^2)$

Differentiate w.r.t. $\hat{X}$

$\hat{X} = \mathrm{E}\{X\}$

$$\mathrm{E}\left\{\left(X-\hat{X}\right)^2\right\} \geq \min_{\hat{X}} \mathrm{E}\left\{\left(X-\hat{X}\right)^2\right\} = \mathrm{E}\{(X-E\{X\})^2\} = \mathrm{var}(X) \geq \frac{1}{2\pi e} e^{2h(X)}$$

where we used the mean of X as the best estimator/predictor for X ($\hat{X} = \mathrm{E}\{X\}$) in MSE sense, and the fact that the Gaussian pdf has the maximum entropy for a given variance. We have equality if $\hat{X}$ is the best estimator, i.e., $\hat{X}$ is the mean of X, and if X is Gaussian

- **Corollary:** Given side information Y and estimator $\hat{X}(Y)$,

$$E\left\{\left(X-\hat{X}(Y)\right)^2\right\} \geq \frac{1}{2\pi e} e^{2h(X|Y)}$$

Replace f(x) with f(x|y)

Achieves the minimum with the conditional mean